

# ACFI315 Finance with Excel

## Lecture 4 Statistical Analysis in Excel



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# Descriptive Statistics

- In order to better understand data, we could conduct statistical analysis in EXCEL.
- The starting point of statistical analysis is to get the descriptive statistics of the data:
  - Mean: the average of a data set.
  - Median: the value in the middle of a data set.
  - Mode: the most frequent value in a data set.
  - Frequency: the number of times that a value occurs in the data set.
  - Minimum: the smallest value of a data set.
  - Maximum: the largest value of a data set.
  - Variance: the degree of dispersion of all data points in a data set.
  - Standard deviation: the square root of variance.
  - Skewness: the distortion of symmetrical distribution or asymmetry in a data set.
  - Kurtosis: the tailedness of the distribution of a data set.
  - ... ..

# Average Function in Excel

- The **AVERAGE** function returns the arithmetic mean of the numbers provided.
- The syntax of the **AVERAGE** function is AVERAGE(number1, [number2], ...).
- Number1 – Required. The first number, cell reference, or range for which you want the average.
- [number2], ... – Optional. Additional numbers, cell preferences or ranges for which you want the average, up to a maximum of 255.

# Median Function in Excel

- The **MEDIAN** function returns the median of the numbers provided.
- The syntax of the **MEDIAN** function is MEDIAN(number1, [number2], ...).
- Number1 – Required. The first number, cell reference, or range for which you want the average.
- [number2], ... – Optional. Additional numbers, cell preferences or ranges for which you want the average, up to a maximum of 255.

# Mode Function in Excel

- The **MODE** function returns the most frequently occurring, or repetitive, value in an array or range of data.
- The syntax of the **MODE** function is MODE(number1, [number2], ...).
- Number1 – Required. The first number, cell reference, or range for which you want the average.
- [number2], ... – Optional. Additional numbers, cell preferences or ranges for which you want the average, up to a maximum of 255.

# Frequency Function in Excel

- The **FREQUENCY** function returns how often values occur within a set of data. It returns a vertical array of numbers.
- The syntax of the **FREQUENCY** function is FREQUENCY(data, intervals).
- data – Required. An array or range of values for which to count the frequencies.
- intervals – Required. An array or range of intervals that you want to group the values in data.

# VAR.P and STDEV.P Functions in Excel

- The **VAR.P** function returns the variance of a population based on an entire population of numbers.
  - The syntax of the **VAR.P** function is VAR.P(number1, [number2], ...).
  - number1 – Required. The first number argument corresponding to a population.
  - [number2], ... – Optional. Number arguments 2 to 255 corresponding to a population.
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- The **STDEV.P** function returns the standard deviation based on the entire population given as arguments.
  - The syntax of the **STDEV.P** function is STDEV.P(number1, [number2], ...).
  - number1 – Required. The first number argument corresponding to a population.
  - [number2], ... – Optional. Number arguments 2 to 255 corresponding to a population.

# SKEW Function in Excel

$$Skew = \frac{n \sum_{i=1}^n (x_i - \bar{x})^3}{(n-1)(n-2)\sigma^3}$$

where  $\bar{x}$  is the mean and  $\sigma$  is the standard deviation.

- The **SKEW** function returns the skewness of a data set.
- The syntax of the **SKEW** function is SKEW(number1, [number2], ...).
- number1 – Required. The first number argument corresponding to a population.
- [number2], ... – Optional. Number arguments 2 to 255 corresponding to a population.



# KURT Function in Excel

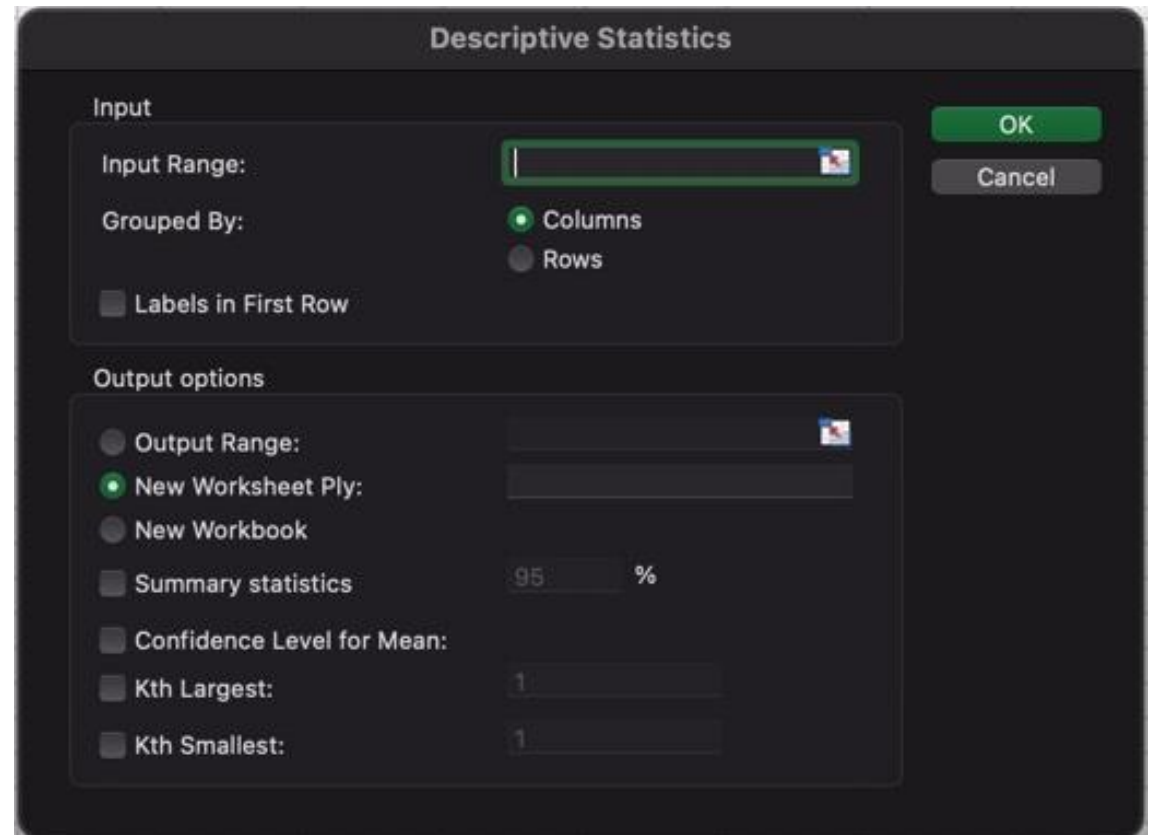
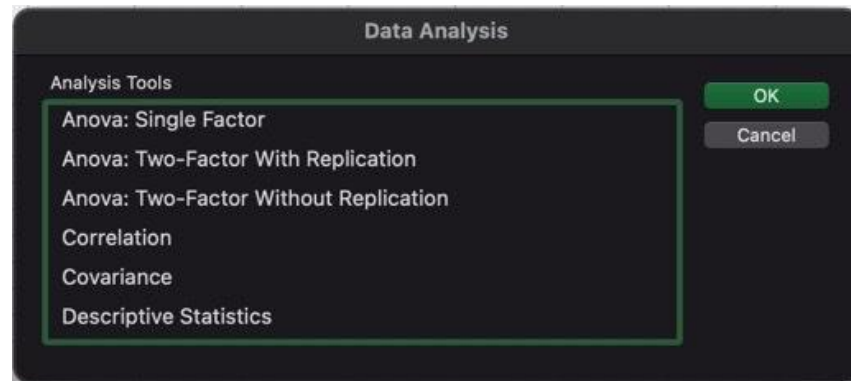
$$Kurt = \frac{n(n+1) \sum_{i=1}^n (x_i - \bar{x})^4}{(n-1)(n-2)(n-3)\sigma^3} - \frac{3(n-1)^2}{(n-2)(n-3)}$$

where  $\bar{x}$  is the mean and  $\sigma$  is the standard deviation.

- The **KURT** function returns the kurtosis of a data set.
- The syntax of the **KURT** function is KURT(number1, [number2], ...).
- number1 – Required. The first number argument corresponding to a population.
- [number2], ... – Optional. Number arguments 2 to 255 corresponding to a population.

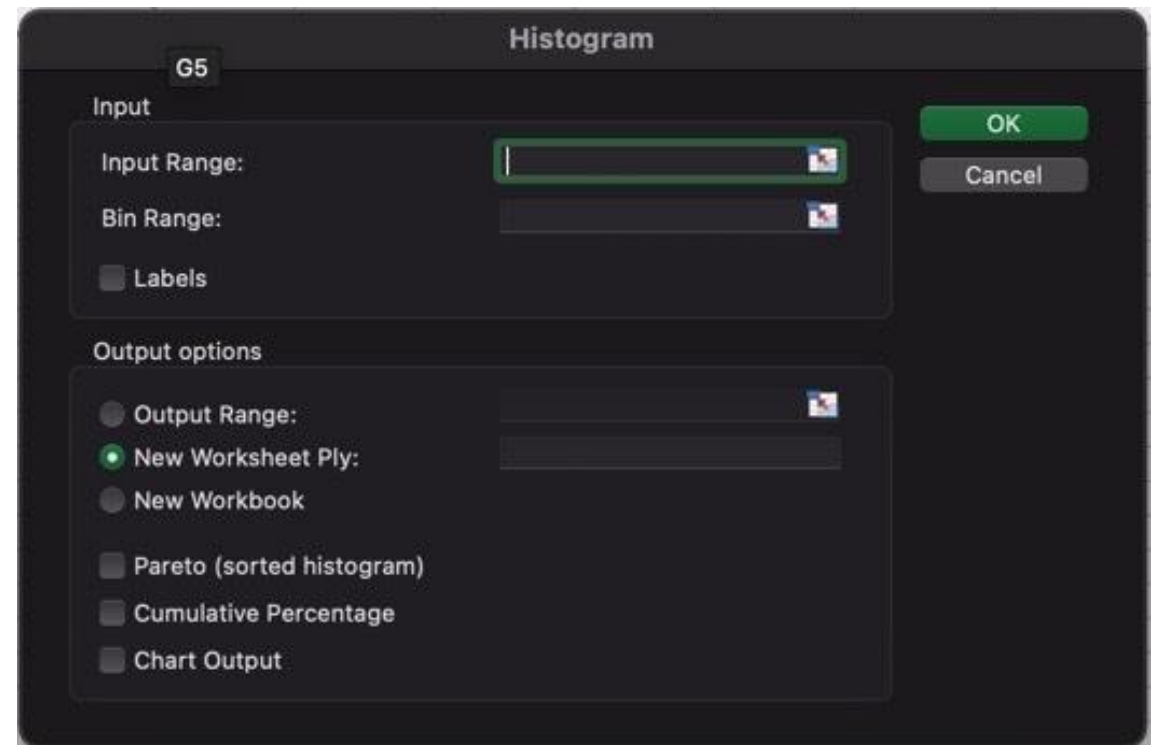
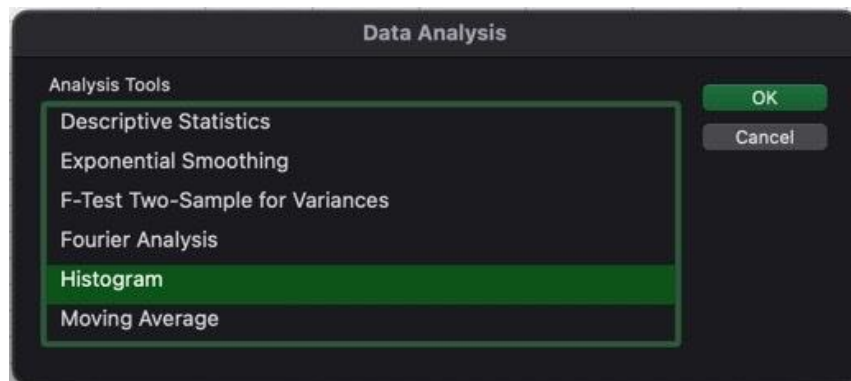
# Data Analysis Tool

- EXCEL has a very sophisticated Data Analysis Tool built in that allows very quick data analysis.
- Data Tab – Data Analysis – Descriptive Statistics.



# Histograms

- A histogram is a graphical representation of data points organized into user-specified ranges.
- Data Tab – Data Analysis – Histogram.



## Other Useful Functions

- ROUNDDOWN(number, digits) returns a number rounded down to a specified number of digits.
- ROUNDUP(number, digits) returns a number rounded up to a specified number of digits.